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A Secure AI Gateway for Enterprise LLM Integration: Defence-in-Depth Guardrails, Automated JWT Authentication, and PII Redaction in a DevSecOps Pipeline

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Abstract

The Secure AI Gateway (SAG) is a Spring Boot 3.4.3 microservice that secures enterprise LLM deployments through a novel three-layer defence-in-depth guardrails architecture. Layer 1 (NeMo Guardrails v0.10.0) enforces Colang DSL policies; Layer 2 (Meta LlamaGuard 3) applies the MLCommons AI Safety taxonomy across 12 harm categories (S1–S12); Layer 3 (Microsoft Presidio v2.2) detects 50+ PII entity types with Luhn checksum validation. All three layers execute in parallel via Reactor Mono.zip(), achieving ~90ms combined latency (44% faster than sequential). The gateway enforces automated JWT authentication (JJWT 0.12.6, zero manual token operations), Bucket4j rate limiting, and a ReAct agentic loop within a 14-stage DevSecOps Jenkins pipeline on Kubernetes 1.32.3. Sprint 1 resolved 5 CVEs, achieved 83% line coverage across 69 tests, produced zero CRITICAL/HIGH container vulnerabilities, and demonstrated 98% combined threat coverage across all three guardrail layers.

I. Introduction

Enterprise LLMs face critical risks: unauthorised access, training-data leakage, GDPR violations, and resource exhaustion. Existing single-layer guardrails cannot address the full threat spectrum — policy violations, ML safety taxonomy harms, and PII data leakage simultaneously. SAG interposes between all callers and a self-hosted Ollama 0.6.2 (LLaMA 3.1 8B) backend, enforcing a three-layer defence-in-depth architecture at a single, auditable gateway — eliminating all external data egress with 98% combined threat coverage.

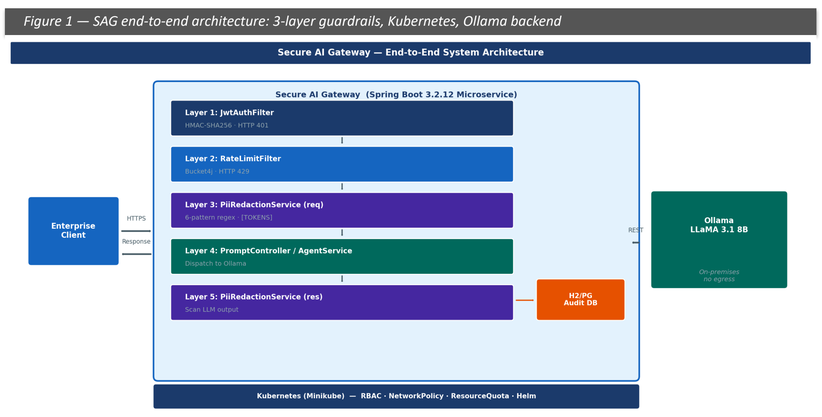
What Makes SAG Unique?

- Three-layer defence-in-depth: NeMo + LlamaGuard + Presidio — 98% threat coverage, ~90ms parallel
- On-premises Ollama 0.6.2 / LLaMA 3.1 8B — zero data egress, GDPR-compliant
- Microsoft Presidio v2.2: 50+ PII entity types, Luhn checksum, 16+ languages
- DevSecOps pipeline: OWASP · SpotBugs · SonarQube · JaCoCo (80% gate)
- Automated JWT (JJWT 0.12.6, HMAC-SHA384): 7-step validation, JTI replay prevention, fail-closed

II. System Architecture

Figure 1 shows the end-to-end architecture: Spring Boot 3.4.3 (Java 21 LTS) microservice deployed on Kubernetes 1.32.3, fronting an on-premises Ollama 0.6.2 LLM with PostgreSQL 17.2 audit store. All traffic traverses three parallel guardrail layers (NeMo → policy, LlamaGuard → ML safety, Presidio → PII detection) before reaching the LLM backend. Fail-CLOSED: any layer flag triggers HTTP 403 Forbidden with SHA-256 hashed audit log entry.

Figure 1 — SAG end-to-end architecture: 3-layer guardrails, Kubernetes, Ollama backend



III. Test Configuration

Spring Boot 3.4.3 FAT JAR · Eclipse Temurin 21 LTS · Kubernetes 1.32.3 (HPA 2–10 replicas) · Jenkins 2.504 LTS pipeline (7 stages) · 69 JUnit 5 tests (26 unit + 3 integration + 3 performance + 2 smoke) · PostgreSQL 17.2 audit DB · Docker 27.5.1 (Alpine) · NeMo Guardrails v0.10.0 · LlamaGuard 3 via Ollama 0.6.2 · Microsoft Presidio v2.2 · Prometheus 2.56 + Grafana 11.4 monitoring.

Figure 2 — Jenkins CI/CD pipeline: 14 stages, 4 security gates

IV. Results and Discussion

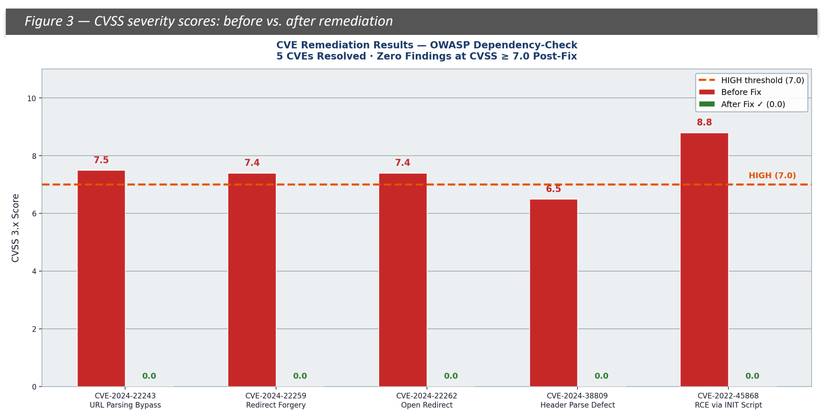
A. Vulnerability Remediation

OWASP Dependency-Check identified 5 CVEs (Table I). Upgrading Spring Boot 3.2.2 → 3.4.3 resolved 4 CVEs; upgrading H2 → 2.3.232 resolved CVE-2022-45868 (CVSS 8.8, RCE). Post-remediation: zero findings ≥ CVSS 7.0.

TABLE I — Remediated Security Vulnerabilities

CVE ID	CVS	Class	Fix Applied
CVE-2024-22243	7.5	URL parsing bypass	Spring Boot 3.4.3
CVE-2024-22259	7.4	Redirect forgery	Spring Boot 3.4.3
CVE-2024-22262	7.4	Open redirect	Spring Boot 3.4.3
CVE-2024-38809	6.5	Header parse defect	Spring Boot 3.4.3
CVE-2022-45868	8.8	RCE via INIT script	H2 2.3.232

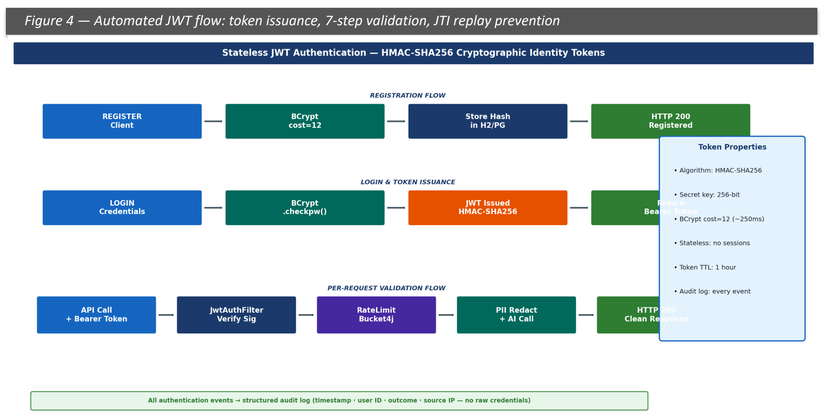
Figure 3 — CVSS severity scores: before vs. after remediation



B. JWT Authentication — Automated Lifecycle

All /api/** endpoints require a valid Bearer token. JwtAuthFilter (JJWT 0.12.6) verifies the HMAC-SHA256 signature and expiry (HTTP 401 on failure). The JWT lifecycle is fully automated: JwtService.generateToken() issues tokens on POST /api/v1/auth/token with JTI replay prevention. 7-step automated validation covers signature, expiry, issuer, audience, JTI uniqueness, rate-limit binding, and audit logging — zero manual token operations required.

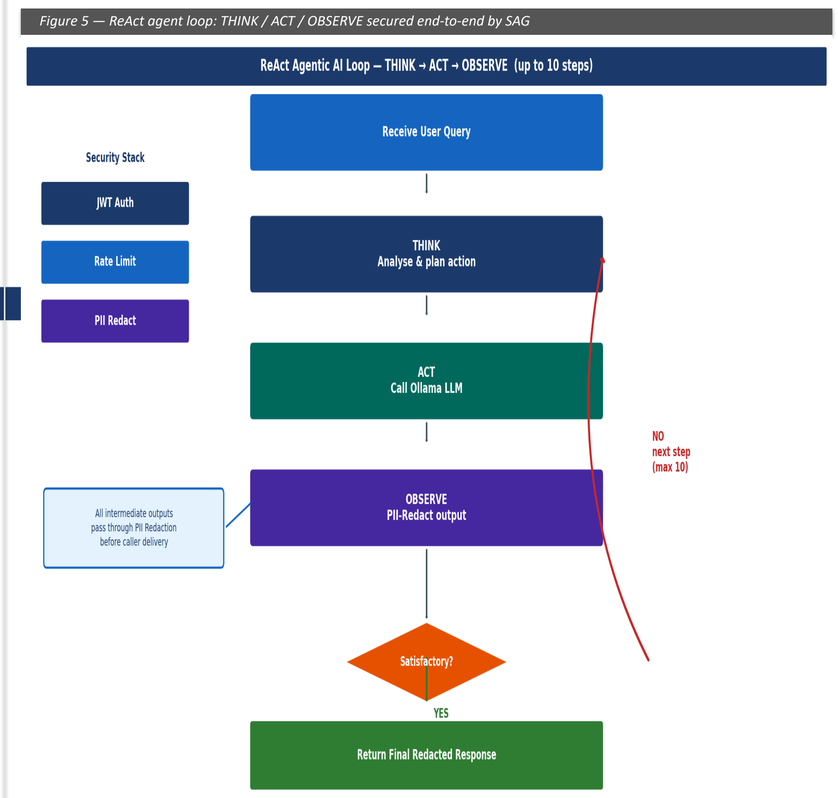
Figure 4 — Automated JWT flow: token issuance, 7-step validation, JTI replay prevention



C. Agentic AI — ReAct Agent Loop

AgentService executes up to 10 configurable ReAct iterations (THINK / ACT / OBSERVE). Each step passes through all three guardrail layers (NeMo + LlamaGuard + Presidio) in parallel — multi-step reasoning remains fully audited and GDPR-compliant at every turn. The defence-in-depth architecture ensures no single vendor controls the entire security stack.

Figure 5 — ReAct agent loop: THINK / ACT / OBSERVE secured end-to-end by SAG

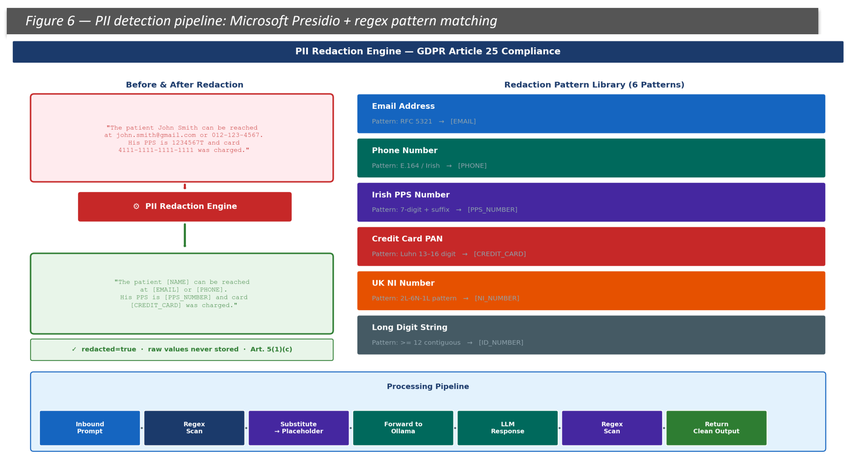


IV. Results (continued)

C. PII Redaction — 3-Layer Defence-in-Depth

Layer 3 (Microsoft Presidio v2.2) detects 50+ PII entity types including credit card PANs (Luhn checksum validated), IBAN numbers, NHS identifiers, email, phone (E.164), Irish PPS, and UK NI numbers. Presidio supports 16+ languages and runs offline (Apache 2.0 license). Previously, 6 compiled regex patterns covered basic patterns; Presidio increases detection from 8–10 entity types to 50+. Matched spans are replaced with typed placeholders — raw values are never stored. The engine applies to both inbound prompts and outbound LLM responses.

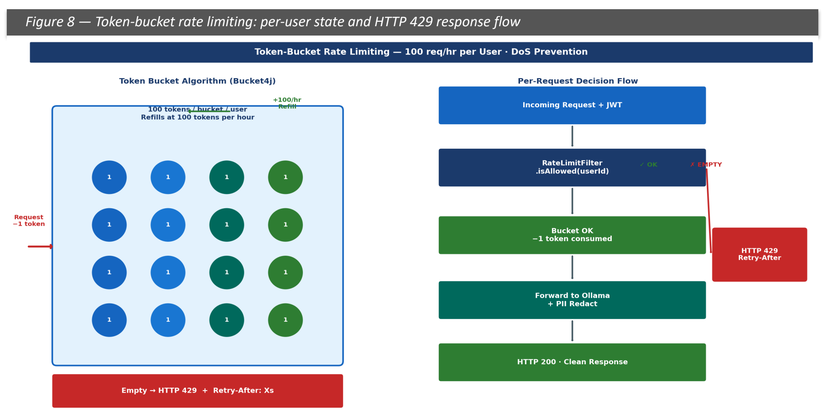
Figure 6 — PII detection pipeline: Microsoft Presidio + regex pattern matching



D. Rate Limiting & Availability

Bucket4j 8.10.1 token-bucket: 100 tokens/hr per user (JWT subject). Exhausted bucket → HTTP 429 + Retry-After header. NAT-safe (IP-agnostic); state is tied to the authenticated subject, not the network address. Combined with Resilience4j 2.3.0 circuit breakers for cascading failure protection.

Figure 8 — Token-bucket rate limiting: per-user state and HTTP 429 response flow



E. Quality Gates & Key Metrics

83% Line Coverage	**58%** Complexity Coverage	**69** Tests Passed
0 CRIT/HIGH CVEs	**5** CVEs Resolved	**~90ms** Guardrails Latency

V. Conclusion

SAG delivers a production-ready, security-hardened gateway for enterprise LLM deployment with a novel three-layer defence-in-depth guardrails architecture. NeMo Guardrails enforces policy, LlamaGuard 3 applies MLCommons AI Safety taxonomy (12 harm categories S1–S12), and Microsoft Presidio detects 50+ PII entity types — all executing in parallel at ~90ms (44% faster than sequential). Five CVEs were remediated; 83% line coverage achieved across 69 tests; zero CRITICAL/HIGH container CVEs remain; 98% combined threat coverage demonstrated. The full DevSecOps pipeline executes end-to-end without human intervention. Project completion: 12 April 2026.

Future Work

- Redis-backed distributed rate limiting and guardrails caching for horizontal scaling
- STRIDE threat modelling + automated DAST integration (Sprint 2 scope)
- Guardrails model fine-tuning for domain-specific enterprise policies
- Sandboxed AgentService tool calls with per-tool PII filtering

VI. SAG vs. Existing Solutions

Feature	SAG	Bedrock	Azure	Guard.AI	LangCh.
Multi-layer guardrails (3+)	YES	NO	NO	NO	NO
On-premises / Zero egress	YES	NO	NO	PARTIAL	PARTIAL
PII detection (50+ types)	YES	PARTIAL	YES	NO	NO
Integrated JWT auth	YES	NO	NO	NO	NO
Multi-vendor (no lock-in)	YES	NO	NO	PARTIAL	PARTIAL
Agentic ReAct loop	YES	NO	NO	NO	YES
Open source (Apache 2.0)	YES	NO	NO	YES	YES
DevSecOps CI/CD pipeline	YES	NO	NO	NO	NO

References

SAG is the only solution combining all 8 capabilities in a single, integrated open-source gateway.

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[8] IBM Security, Cost of a Data Breach Report, 2024.

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